Break It 'Til You Make It
An Exploration of the Ramifications of Copyright Liability Under a Pre-training Paradigm of AI Development

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ABSTRACT

This paper considers the potential impacts of a pre-training regime on the application of copyright law for AI systems. Proposed evaluations of the use of copyrighted works for AI training have assumed a tight integration between model training and model deployment: the model's application plays a central role in determining if a training procedure's use of copyrighted data infringes on the author's rights. In practice, however, large, modern AI systems are increasingly built and deployed under a pre-training paradigm: large models may be trained for a multitude of applications and then subsequently specialized toward specific ones. Thus, I consider copyright's indirect liability doctrine to consider the effect of copyright on the current market structures involved in the development and deployment of AI systems. The main contribution of this paper lies in its analysis of what indirect copyright liability litigation for technologies in the past have to say for how AI companies may manage or attempt to limit their copyright liability in practice. Based on this analysis, I conclude with a discussion of strategies to close these loopholes and of the role that copyright law has to play within the AI policy landscape.

CCS CONCEPTS

• **Social and professional topics** → **Secondary liability**; **Copyrights**; • **Computing methodologies** → *Learning paradigms.*

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1 INTRODUCTION

Recent progress in the development of generative AI systems has brought questions of copyright liability to the attention of regulators, scholars, artists, technologists, and the general public alike. This technology facilitates the creation of new works at the same time as it threatens the livelihood of artists whose works are used in training.

The legal literature has focused primarily on whether a model's use of copyrighted works as training data directly infringes upon a copyright holder's rights. While questions around direct liability from model training on copyrighted works are far from settled, a common way an answer has been attempted in the legal literature relies on the conceptualization of a close connection between the purpose of model training and deployment application. I argue that this reasoning and conceptualization warrants a closer look. In particular, it is out of step with the *pre-training* process that has emerged as the dominant paradigm for model development [Bommasani et al. 2021; Devlin et al. 2018; Ramesh et al. 2022]. Large models are usually trained with generic tasks (e.g., next-word prediction) on broad datasets and *fine-tuned* (i.e., re-trained) on narrower and, typically, smaller task-specific datasets. Furthermore, pre-training is expensive and creates a split in the market for these systems between a small group of high-resource corporate actors [Microsoft 2023; OpenAI 2019] that produce pre-trained models and a large group of smaller downstream actors that *fine-tune* pre-trained models [Bender et al. 2021]. This means that a single training process is often subject to a wide variety of uses by multiple actors.

Proposed tests for direct liability adopt a conceptualization of the training process that may not align with the technical reality of the training process or the market structures surrounding AI development. With the introduction of multiple actors during the AI development and deployment process, I consider the implications of indirect liability. Indirect copyright liability is a judicially developed provision of copyright law that potentially extends infringement exposure to developers of devices or services that contribute to, induce, or vicariously benefit from downstream infringing activities [Menell et al. 2023]. Generative AI can facilitate output or model infringements by downstream developers [Lu et al. 2019]. I draw an analogy between indirect liability from pre-trained models and indirect copyright litigation involving peer-to-peer file-sharing networks to identify opportunities for model developers to subvert policy goals. I highlight these strategies to illustrate how the "breaking" of the AI development chain within a pre-training paradigm could be further exploited, allowing involved players to "make it'– to bypass liability for the use of copyrighted works. In line with scholarship that emphasizes accountability at each stage of model development [Khan and Hanna 2023], I propose duties that may

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apply during key stages of model development: training and deployment. I conclude with a discussion of copyright's role within the AI policy landscape.

2 PROPOSED TESTS FOR EVALUATING THE USE OF COPYRIGHTED WORKS IN MODEL TRAINING ARE NOT ATTUNED TO THE STRUCTURE OF THE AI DEVELOPMENT MARKET.

2.1 Proposed analyses for direct infringement rely on a conceptualization of the developmental process as a *single* training process toward a *single* application.

Proposed evaluations in the literature of the legality of the use of copyrighted works to train AI models heavily rely on model application and model outputs, and hinge on analysis through copyright's protection of expression. As part of these evaluations, the developmental process and technical architecture is conceptualized as a single pipeline–a single entity that builds a single model toward a single application. Indeed, model training is conceptualized as a process that learns toward a particular application.

Sobel [2017] posits that the use of copyrighted works to train facial recognition algorithms does not implicate copyright protections: "Training facial recognition algorithms on copyrighted photographs does not implicate the works' protectable aspects. The use analyzes factual information—the unique physical features of a subject's face—in the photographs, rather than photographers' expressive choices." As part of this analysis, a machine learning model that recognizes faces–a non-expressive application– is conceptualized to take only from the non-expressive aspects of the copyrighted works in the training process. Since copyright law protects only an author's original expression, this non-expressive use of copyrighted works then falls outside the scope of copyright protections and is non-infringing. This framing is reflected throughout the legal literature. Quang [2021] explains that ML models that mine data for "functional, non-expressive purposes" extract "unprotectable ideas and patterns from data" as part of their training process. Sag [2018] also reflects this sentiment, noting that "using software to scan texts to identify patterns of speech or unusual relationships between words" would be considered non-expressive use. The safe harbor for certain non-expressive, digital uses of copyrighted works is additionally reflected in legal precedence such as *Authors Guild v. Google*, 721 F.3d 132 (2d Cir. 2015); *Perfect 10, Inc. v. Amazon.com, Inc.*, 508 F.3d 1146 (9th Cir., 2007).

But things change when the model's resulting application is deemed expressive. Lemley and Casey [2020] write: "Some ML systems will be interested in the expressive components of the work as an integral part of their training. That is, the goal will be to teach the system using the creative aspects of the work that copyright values, not just using the facts or the semantic connections the law is not supposed to protect. That is particularly likely of those systems like MuseNet that are training in order to generate their own expressive works". Again, the system is conceptualized as one that learns specifically from a singer's creative expression in their song toward the production of a model that generates a song similar to theirs–one model, one developer, one application. Because the artist's expression is implied in this use, this then triggers potential liability under copyright law subject to a fair use defense.

2.2 A pre-training paradigm of AI development connects a *single* training process with a *multitude* of downstream applications.

In the previous section, I discuss how proposed evaluations of AI model training rely on a conceptualization of the development process as a single pipeline toward specific applications. However, this conceptualization does not match the current machine-learning ecosystem and supply chain. The path from a model's training process to a resulting system's output may no longer be so straightforward. Under a pre-training paradigm, high-resource developers create *pre-trained models* that can be adapted for a wide range of downstream applications. Under the technical conceptualization embedded in challenge for copyright enforcement is the fact that the same training process can inform both expressive and non-expressive applications. At the same time, with the use of pre-trained models that may be directly trained using copyrighted works, downstream developers and deployers may interact with little to no protected data.

A pre-trained model is initially trained on a broad dataset with a generic training objective. For example, natural language processing (NLP) systems, such as OpenAI's ChatGPT and Anthropic's Claude, are initially trained on broad, internet-scale datasets to model the data (e.g., predicting a hidden or masked portion of a document, given preceding and following words) [Brown et al. 2020]. These models are then *fine-tuned* for specific tasks. Fine-tuning typically requires a relatively small amount of specialized data but can also occur with no additional data or training. For example, Jaques et al. [2016] demonstrate the use of reinforcement learning to fine-tune recurrent neural networks for music generation through a data-free application of reinforcement learning and music theory. In NLP systems, systematic changes to a model's input, such as appending a user's input to a fixed set of instructions, can adapt a model to downstream tasks with zero additional training [Brown et al. 2020]. This process is often called *prompt engineering*.

Pre-training requires a lot of compute and data. This currently limits the production of these models to high-resource developers [Dotan and Milli 2019]. As a result, a single pre-trained model is often used for a variety of AI systems. This creates challenges for legal tests that rely on a tight connection between model training and model deployment. Consider, for example, the classic ImageNet dataset [Russakovsky et al. 2015], which labels a wide range of images with object categories. Features learned by training a model on this generic classification task are effective for a variety of downstream tasks that range from object detection and classification to image generation [Huh et al. 2016]. This demonstrates how a model produced by a single pre-training process can be fine-tuned toward both recognizing stop signs to improve the safety of autonomous vehicles (a non-expressive application discussed in [Lemley and Casey 2020]), as well as potentially infringing and expressive applications, such as generating artistic visual images.

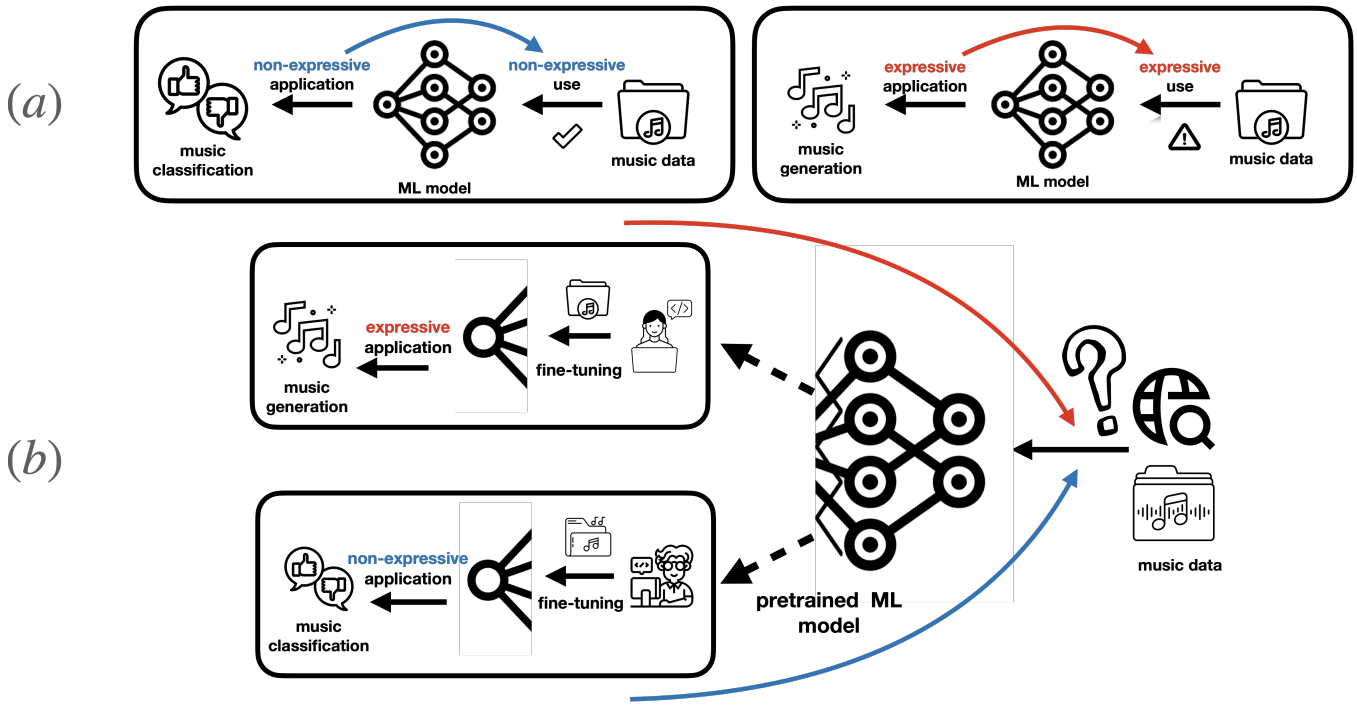

Figure 1: (a): This is a representation of the conceptualization of training in proposed analyses for AI systems in the legal literature, which have tended to focus on expressive/non-expressive use. (b): This conceptualization could lead to complications under a pre-training paradigm of AI development. Fine-tuning involves training the last layer of the pre-trained network toward a specialized task. Pre-trained models produced from training on copyrighted works can be applied in ways that have been characterized as both expressive and non-expressive. Pre-trained models may not always embed any eventual applications or outputs, potentially breaking the relationship that is centered in proposed judicial tests to determine whether the use of copyrighted works in training models is expressive or non-expressive.

2.3 Illustrative Example: Music Generation

In this section, I provide an example on music generation leveraging proposed evaluations to show complications arising from how its conceptualization of the training process applies to current market structures associated with AI development. Consider a music generation system that uses a pre-trained music model, such as in [Agostinelli et al. 2023] or [Zeng et al. 2021]. This pre-training step uses a large dataset of sound recordings that includes, among others, Ariana Grande's catalog. Zeng et al. [2021] show how such a model can be repurposed effectively for both expressive (melody completion) and non-expressive (accompaniment suggestion, genre classification, and style classification) tasks.

The model in [Zeng et al. 2021] is fine-tuned for melody completion with examples that are labeled as good and bad completions of the melody. We will suppose that this step does not use any of Ariana Grande's catalog, as the datasets for these tasks are typically much smaller[1]. Finally, the model is deployed and produces a viral melody completion that is substantially similar to Ariana Grande's song "Thank U, Next" and a legal battle ensues.

In the scenario, it is plausible that the use of Ariana Grande's music to train the initial model would result in a finding of copyright infringement. Ariana Grande can prove that the defendant actually copied "Thank U, Next" and that the melody produced by the AI system is (by assumption) substantially similar to the presumably protected work. The answer to the question could become more complicated with pre-training and fine-tuning split across different entities. For example, suppose that Company A pre-trains an "embedding" model[2] that represents songs as high-dimensional vectors. Separately, Company B fine-tunes a generative layer on top of these embeddings based on a small set of public domain examples of good and bad melody completions. The final system is the same as before, as is the potentially infringing output.

However, in this case, both companies could have a good defense against direct copyright liability. Company A produced a model based on a generic dataset. It is hard to argue that this is copying an artist's expression. Similarly, Company B only used public domain data that conveyed generic information about good and bad melody completion. It is even possible that none of Company B's employees have listened to Ariana Grande's music. Because the purpose of use changes at different steps of model training, the attribution of liability becomes less obvious. Furthermore, the most defensible

[1]We could also suppose that the company lawyers have required that engineers use music in the public domain in this step to reduce liability.

[2]see, e.g., Boykis [2023] and https://platform.openai.com/docs/models/embeddings.

phase of training (pre-training) may also be the phase that actually uses copyrighted data.

3 THE GULF BETWEEN TECHNICAL CONCEPTUALIZATION AND DEVELOPMENTAL REALITY CREATES REGULATORY UNCERTAINTY.

The technical utility of pre-training and then fine-tuning is well established [Devlin et al. 2018; Russakovsky et al. 2015]. From the perspective of proposed direct copyright liability tests, this paradigm of development could also be favorable to corporations with the resources required to pre-train high-performing models. Moreoever, copyright's constitutional roots in a consideration of technological progress and welfare could support the claim that the production of pre-trained models is a non-infringing use of copyrighted works. For example, Quang [2021] discusses how a pre-training paradigm can support innovation through increased access to AI system development for smaller developers that build upon pre-trained models.

The combination of multiple actors and training processes could complicate the attribution of copyright liability, and rightsholders may additionally turn to *indirect liability* and *indirect infringement* as a way to protect their work. Indirect liability applies to technology that facilitates downstream copyright infringements. From a practical perspective, indirect liability suits are important for enforcement because it can be infeasible or impractical to target a diverse and distributed population of direct infringers [Menell 2013]. However, I identify ways that the application of indirect copyright liability to the modern ML supply chain may be evaded. First, in certain instances, a primary or direct infringer in the sense of copyright law may simply not exist. Second, the Sony Safe Harbor may afford developers of pre-trained models with a strong defense against claims of indirect infringement.

3.1 Identifying direct infringement could be complicated at multiple points.

Applying indirect copyright liability starts with determining direct infringement–without direct infringement, there is no indirect infringement [Menell et al. 2023]. In the case of pre-trained models, it may be difficult to establish direct infringement from model fine-tuning or use[3]. The complexity of the interaction between model pre-training, fine-tuning, and use is difficult to disentangle. Furthermore, downstream users and developers often have limited knowledge about training data or mitigation techniques.

Fine-tuning. One possible location for liability in the generation of infringing output is with the model fine-tuner. This is illustrated as part of Figure 1 when the downstream developer re-trains their model to produce popular music that matches a particular style and genre. The fine-tuner's choices change model output and may increase the likelihood that output infringes on protected work from the original training set. One could argue that deploying this fine-tuned model infringes on the protected work used in pre-training. However, this is a tenuous connection because the fine-tuner may not have direct access to or awareness of the relevant protected work and the fine-tuning may involve little or no protected work directly.

Model use. Another possible actor who may directly infringe on work in the training set is the model user who prompts a model for output. In some cases, it may be quite clear that these prompts cause an infringement. For example, prompting the model from 2.3 to "generate a song like Ariana Grande's 'Thank U, Next' that is about cats" may make it easy to hold a user liable for direct infringement. This is supported by a potential analogy between generative models and databases, where the model essentially acts as a storage mechanism for the training data. For example, [Henderson et al. 2023] finds that, with certain prompts to pre-trained language models, a user may be able to leak an entire book or poem from a pre-trained language model.

However, users may have little understanding of the underlying models and data. They may be unaware that models have the ability to reproduce training data. Alternatively, they may prompt a model in a way that merely happens to generate infringing output. For example, a user may prompt a model to generate a song that is in the pop genre with a lead singer and a catchy melody — i.e., unprotected features of 'Thank U, Next' — and get back an infringing output. In this case, a user may argue that either the fine-tuners or pre-trainers caused the infringement, as there are many non-infringing ways to respond to this prompt.

3.2 The Sony Safe Harbor could further shield developers of pre-trained models.

Even after direct infringement is established, model pre-trainers may be able to claim the *Sony Safe Harbor* to shield themselves from liability. This safe harbor was developed in [*Sony Corp. of America v. Universal City Studios, Inc.*, 464 U.S. 417 (1984)] as part of the doctrine of contributory infringement. In [*Sony Corp. of America v. Universal City Studios, Inc.*, 464 U.S. 417 (1984)], rights holders to television programming sued Sony for downstream infringing uses of Sony's video recording technology. The courts did not hold Sony liable under the rationale that **the developer of a dual-use technology should *not* be held liable for contributory copyright infringements if their technology could also be used toward *substantial, non-infringing uses.*** This was justified by a precedent from patent law and the potential public benefit of the technology [Menell and Nimmer 2007]. This safe harbor played a critical role in the resolution of a string of copyright cases that involved peer-to-peer (P2P) file-sharing networks.

To see how this safe harbor could apply to the developers of pre-trained models, let us return to our example from Section 2.3. The pre-trained music model in question produces meaningful, machine-readable representations of songs. These representations, or embeddings, can be used for multiple tasks that range from song classification to completion. The developers of pre-trained models could argue that their systems serve many possible uses and thus claim the Sony Safe Harbor.

This may be reinforced by the stated goal of machine learning companies to produce "general" models. For example, in computer vision systems, pre-trained embeddings may be used downstream

[3]This is in contrast to indirect liability litigation involving peer-to-peer file-sharing networks where establishing direct infringement was relatively straightforward because of direct interfacing with copyrighted works.

to improve the safety of autonomous vehicles by better recognizing stop signs. Similarly, natural language processing systems may leverage pre-trained embeddings to identify and/or reduce bias[4] and the prevalence of hate speech on online platforms. At the same time, these embeddings can clearly support generative applications that infringe on the training data. Based on the precedent set in P2P file-sharing, even if non-infringing use is a relatively small fraction of use in practice, the Sony Safe Harbor may protect the use of copyrighted works for pre-trained models from claims of indirect infringement. Balancing the potential benefits of a pre-trained model against the potential for infringing use is complicated. However, as I discuss in Section 5, an understanding of how a model is used downstream is a prerequisite to an informed decision. s

4 EXPLOITING GAPS IN COPYRIGHT ENFORCEMENT, DEVELOPERS AND DEPLOYERS OF GENERATIVE MODELS MAY TAKE DIFFERENT APPROACHES TO THE RESEARCH, DESIGN, AND DEPLOYMENT OF GENERATIVE AI SYSTEMS.

Helman [2010] discusses how the open-ended indirect liability doctrine in the Sony standard, as a result of the subsequent P2P cases, has induced dual market responses in subsequent technologies. I point out the potential for a similar response by market players in the ML ecosystem. On the one hand, there are well-resourced incumbent model developers. These companies often seek to profit directly from hosted services and APIs. This creates an incentive to develop risk management methods that allow them to claim the Sony safe harbor while profiting off of downstream fine-tuning and prompt engineering applications. On the other hand, there is a growing ecosystem of non-incumbent developers that aim to profit from the open-source ML ecosystem. These actors profit indirectly from model development and have an incentive to release models in a way that limits knowledge or control of downstream use.

In this section, I consider market responses from both groups to indirect infringement doctrines. I split our analysis along three dimensions: choice of service disaggregation, choice of model release, and choice of R&D investment. The common thread tying together responses from both groups of actors in the ML ecosystem to the enforcement of copyright law may be that everyone wants to know enough to have their slice of the pre-training pie, but no one wants to know enough to be held liable for potential infringements.

Choice of service disaggregation

One direction that model developers may take to limit liability is through the *disaggregation of services*, where a technology pipeline is split amongst different actors in a way that makes it hard to attribute liability. When one actor sources data, trains a model on that data, and then leverages that model toward an application, pinpointing which actor to assess the copyright liability of is straightforward[5]. On the other hand, when the technical and organizational pipeline is disaggregated, the determination of infringement *and* how each actor is evaluated in light of infringement is complicated. The example in Section 2.3 shows how disaggregating the pre-training and fine-tuning of a model complicates the assessment of liability for both actors. Company A, who pre-trained the model, can claim the Sony Safe Harbor, while Company B, who fine-tuned the model, can claim that Company A's actions caused the infringing output. Adjudicating these claims will challenging and likely relies on the details of a complex technical system.

Disaggregation of technology production to limit liability is not new to technology development generally or indirect copyright liability specifically. For example, Nissenbaum [1996] discusses how the lack of a culture of accountability combines with the problem of "many hands" to complicate liability attribution in a computerized society. With respect to indirect copyright liability, disaggregation of services in peer-to-peer (P2P) file-sharing networks was a key strategy for technology developers to limit liability. Choi [2005] discusses the challenges that disaggregation presents to determinations of indirect copyright liability in the wake of *MGM Studios, Inc. v. Grokster, Ltd.*, 545 U.S. 913 (2005).

This calls for attention to how services are disaggregated in ways that can subvert liability in an ML ecosystem. Furthermore, note that disaggregating pre-training and fine-tuning is not the only way to divide the ML supply chain. For example, consider the emergence of *data laundering* as a strategy [Jiang et al. 2023]. Rather than collecting data directly, companies can fund non-profit or academic organizations to curate datasets and distribute them under permissible licenses. While data laundering allows potential competitors to access the data, it can make it more difficult to attribute credit for the use of copyrighted works as part of model training. As the ML supply chain matures, it is reasonable to expect companies to look for ways to maximize profit while minimizing liability. In order for regulation to keep pace with technology development, it will be important to recognize that disaggregation is often a design choice made by model developers and deployers. Appropriately assessing liability may require that courts understand the intent behind such choices and reasonable alternatives a developer could have made–namely, beyond an inquiry into where abstraction boundaries *exist*, an understanding of where they are *created* to shield from liability [Selbst et al. 2024].

Choice of model release

Solaiman [2023] lays out a variety of different ways to deploy generative AI systems. These range from restricted releases, such as gradual or staged releases, hosted access, and cloud-based models, to freely available and downloadable models. The crucial distinction from a copyright liability perspective is that some model releases allow operators visibility and potential control over their downstream uses. For example, model use through a hosted API might be limited to probing inputs/outputs rather than changing model weights. Similarly, a model that is hosted on the cloud gives operators the capacity to monitor the operations that are sent to the model by the user. The amount of visibility and control that model releases permit have relevance to how model operators are assessed under copyright law.

For example, the Ninth Circuit Court in Napster's case held that Sony's substantial non-infringing use only extended to cases where companies had *constructive knowledge* of infringement, not *actual knowledge* of infringement. Constructive knowledge is the knowledge that one *should* have, the knowledge that is reasonable for one

[4] [Levendowski 2018], e.g., advocates for the use of copyrighted works toward this purpose.

[5] This is not to say that the application of copyright law will be trivial, but that there is only one actor whose liability is evaluated under copyright law.

to have, whereas actual knowledge is the knowledge that one *does* have. Once companies have actual knowledge of an infringement, they are no longer protected under the safe harbor [Fitts 2006]. Lee et al. [2023] discusses how the visibility that hosts have into the operations that users are sending to the models could also translate into greater expectations under copyright law about the degree to which developers monitor and control infringing/non-infringing uses of their technologies. For example, if a user queries a model to send embeddings of Ariana Grande's music, it could be possible for the host of the model to prevent such queries and, thus, be liable for the failure to do so.

As a result, doctrines of indirect infringement may create an incentive to release models in a way that limits a developer's knowledge about or control of subsequent model use. For non-incumbent players, who might seek to build off the ecosystem that a model release supports, this may create incentives for models to be released as downloadable and executable code[6]. For incumbent players that try to monetize model use directly, this incentive is counteracted by the value of user data in improving models. However, they will face an incentive to leverage this data in ways that minimize actual knowledge of misuse.

Choice of R&D investment High-resource developers of pre-trained models might have an incentive to control the model ecosystem and to protect their trade secrets. Monitoring uses of models is integral to improving them. However, the level of control and knowledge that developers have over how users interface with models could also mean increased liability under doctrines of indirect infringement.

As a result, model developers may invest in research that allows them to train on data from model use without actual knowledge of the nature of use. For example, developers could invest in privacy-preserving training methods that leverage homomorphic encryption [Chen et al. 2018]. Such methods would allow a model to train on data that the developers can not access or observe. This has some precedent in P2P cases where privacy-enhancing technologies were leveraged in multiple attempts to limit liability–through decentralizing computation and encrypting transfer files [Choi 2005].

Separately, we may see strategic investment in guardrails to support other company goals. Incumbent players may develop costly guardrails as a competitive strategy to create barriers to entry for competing developers. If costly mitigation measures become an industry standard, then this makes it hard for new players to enter the market. Similarly, guardrails that filter input or output (e.g., disallowing prompts that generate art "in the style of" a particular artist or a variant of ContentID [Popper 2016] to filter output) may create legal hurdles for actors that seek to release models directly because such guardrails are hard to include or easy to bypass in open releases.

5 DUTIES

Given the ways that liability can be evaded under a pre-training paradigm of development, I propose considerations for evaluations of copyright infringement. In particular, I suggest that tests for indirect copyright liability for model developers should embed a consideration of whether developers of ML models operate with a reasonable duty of care [7] **in monitoring data sourcing and influence** and **combatting infringing model applications**–this could include not just evaluations of infringement given the model at-hand but more global evaluations of potential downstream uses of pre-trained models to build generative models that infringe on authorial works [Paullada et al. 2021]. Rather than impose strict technical or developmental requirements or outcomes, this duty would require a reasonable effort commensurate to factors such as a company's size or a person's qualifications.

This, again, is not an entirely novel proposal for indirect copyright infringement tests. Picker [2004] suggests an "ongoing design obligation"– a modification of the Sony test to condition the safe harbor upon a design of technologies that are able to "phone home"–networked technologies. Namely, "the producer should face a substantial non-infringing use test, coupled with the duty to evolve the product to eliminate infringing uses" [Picker 2004]. Unlike the VCR and other technological devices, the technologies that we interface with (arguably including pre-trained models) are increasingly networked–product release is not the end of the story. Rather, technological products can and should be expected to "phone home" and iterate. Design choices that have implications for copyright liability are made at each step of developing models and even after models are released. Like Selbst et al. [2024] suggests, courts can and should open these design decisions up. Core to both Picker [2004]'s suggestion and ours is that, rather than subjecting a technology to an all-or-nothing one-time evaluation, judicial tests and policies can provide incentives for ongoing technological innovations that prevent infringement. This is in keeping with the constitutional basis for copyright law, which emphasizes both technological and scientific progress alongside authorial incentives to create.

Training Duties. As part of training duties for ML systems, it is important to look into what datasets contain and how they are labeled. And, perhaps, beyond an understanding of what is ingested, how copyrighted works are ingested as part of the training process to influence resulting system applications and outputs. This could be done, for example, through the use of attribution models, such as [Dai et al. 2023]. This could lend to more preemptive attribution and recourse for the authors of copyrighted works. Proposed EU copyright legislation has included a requirement to disclose copyrighted content used to train ML models [Vincent 2022]. And, in [*Authors Guild v. Google*, 721 F.3d 132 (2d Cir. 2015)], a large part of the transformative use determination of Google's data mining came down to the snippets in Google Books tracing back to author works, giving authors an opportunity to profit from the technology. By attributing copyrighted works used in the pre-training process, authors have an opportunity for their works to be discovered through this technology. This may allow artists to benefit from a model developer's use of their work and support the development of licensing regimes for data use. This could also recalibrate fair use considerations that center on market harm.

However, unlike technologies that have been investigated under indirect infringement in the past, copyrighted works also contribute

[6] I.e., to release models as "devices" rather than "services."

[7] This approach is inspired by Danielle Citron's proposal that a duty of care to prevent "cyber harassment, cyberstalking, and intimate privacy violations" apply to internet platforms in order to receive liability protection under Section 230 [Citron 2022]. This is arguably an analogous situation where technology facilitates behavior with established harms where it is difficult under U.S. law to hold perpetrators accountable.

to the inherent technical functionality of ML systems. ML systems could present a challenge to legal frameworks for analyzing technological fair use and copyright liability that categorize these technologies' use of copyrighted works as creational, operational, or output uses.[8] Copyrighted works do not merely serve as raw material in the creation or operation of these technologies, but the work also "in theory, becomes part of an output to end users" [Gillotte 2019].

Individualistic conceptions of data and expression in copyrighted works can obscure the value that is gained through data in the development of AI systems. This makes the role and value of data that is ingested as part of AI training different from data that is hosted on P2P networks. Namely, the act of downloading a popular song from the Billboard Top 10 does not necessarily implicate other songs. On the other hand, when model deployers and fine-tuners want to build a model to generate work in the style of an artist, unexpected works in the training data may be found to contribute to the end result. Furthermore, at the scale of billions of data points, any *one* piece of data (or, *one* copyrighted work) will likely do little to change the functionality–including the generative functionality–of the model. This makes a technical focus on guaranteeing non-expressive or non-similarity properties of model outputs while non-consensually training on copyrighted works potentially neglectful of the foundational role of copyrighted works as part of datasets. In particular, an author's expression may not only be present in her work but also in aspects of the works of others–and those aspects could be specifically technically tagged to draw from in the production of similar works. To further lay out duties for data collection and use for model training, it may be helpful to look to recent proposals that recognize the social and collective *value* of data, such as [Gordon-Tapiero et al. 2022; Parsons and Viljoen 2023] that build on and reform privacy and data protection, which have both historically focused on the process surrounding data collection and use.

Deployment Duties. At deployment time, duties could prevent model developers from profiting from the deployment of systems that interfere with the market for the original work without the associated copyright liability. Menell and Nimmer [2007] discuss how the legislative history of copyright law points toward tort principles and suggests that the tort framework could better balance innovation and copyright protection: if a technology can be designed in a different way to prevent infringements, it should be. This could involve the deployment of technological guardrails: system designers may have a duty to identify and deploy technological guardrails to prevent users from creating infringing outputs (e.g., preventing users from using prompts that ask for work to be output in the style of a particular artist).

The potential for infringing uses and outputs of deployed ML systems are particularly clear for pre-trained generative art and text models where barriers to entry are low. Users can input plaintext prompts to generate (or, perhaps, retrieve [9]) works that are highly similar to existing works by artists[10]. In such cases, governing the inclusion of artists' works in the training set of pre-trained models may not be enough. They could be supplemented by steps to monitor or and evaluate how artists' expression could be elicited as part of downstream uses. The expectation of monitoring for downstream harms may be especially important for pre-trained models that could eventually have generative model outputs–and especially where certain kinds of use cases could produce outputs that directly compete with artists' works–even if they may not regurgitate or necessarily plagiarize those works. As part of a complaint against StabilityAI for indirect liability [*Anderson et al. v. Stability AI*, (N.D. Cal. 2023) (No. 3:23-cv-00201)], the count states:

> Individuals have used AI Image Products to create works using the names of Plaintiffs and the Class in prompts and passed those works off as original works by the artist whose name was used in the prompt. Such individuals are referred to herein as 'Imposters' By using a particular artist's name, Imposters can cause the AI Image Product to rely more heavily on that artist's prior works to create images that can pass as original works by that artist. These output images are referred to herein as 'Fakes.

Monitoring and mitigating how models are applied to authors of creative works could strengthen opportunities for compensation and mitigate market harms that current developers enable.

6 CONCLUSION AND AI POLICY IMPLICATIONS

Simply scraping an image or building a technical system that creates an output that draws from a copyrighted work might not automatically trigger copyright infringement[11]. Rather, it is the causal process of copying between these events that gives rise to potential liability. By introducing multiple training processes, intermediary developers, and additional network layers, a pre-training paradigm of machine learning challenges the legal conceptualization of this causal process. It can render both the responsibility of those involved as well as the technical link between data and output tenuous. In this paper, I outline potential ramifications copyright liability under a pre-training paradigm of AI development. First, I evaluate pre-trained models under currently proposed judicial tests for direct infringement in the use of copyrighted works for the training process. I show how these tests rely heavily on a conceptualization of ML systems and the development process as a single pipeline with one model and one application. A pre-training paradigm breaks this presumption and challenges these tests with the introduction of a single training process informing a multitude of downstream applications. Given the role of pre-trained models in facilitating the development of downstream models, I analyze a pre-training paradigm under a indirect copyright liability analysis. I discuss why indirect liability can be difficult to establish and discuss lessons for pre-training based on how judicial tests for indirect copyright liability have been subverted in the past. Finally, in light of these strategies, I discuss duties of care as an approach to regulating the developers of pre-trained models.

[8] See [Lee 2009], which proposes a framework for analyzing fair use of different kinds of technologies, categorizing uses into creational, operational, and output uses.
[9] See, e.g., experiments in [Henderson et al. 2023]
[10] See, e.g., https://twitter.com/Kelly_McKernan/status/1651749436358291460 and [Casper et al. 2023].
[11] As Balganesh [2012] notes, for copyright law: "Neither the bare act of reproduction nor the mere production of a substantially similar work is sufficient to trigger liability without the other."

I conclude with a few remarks on the limitations of our analysis and on implications for AI policy as a whole. The judicial application of copyright law does not have to be all-or-nothing –injunction or the status quo. Ginsburg [2001] notes that "the copyright statute and the judges who interpret it attempt a balance: Creators should maintain sufficient control over new markets to keep the copyright incentive meaningful, but not so much as to stifle the spread of the new technologies of dissemination." Perhaps, access to emerging generative models and AI technologies is a public benefit. The harm does not so much stem from the existence of these technologies but the infringements that occur from the development or the use of these technologies. As part of legal remedies that have come from copyright cases involving technologies, plaintiffs have been more successful in receiving compensation when they participate in the market for the technology, such as through compulsory licensing mechanisms or through the monitoring of infringements when compared to a complete injunction of that technology [Ginsburg 2001]. Copyright suits could pave the way for re-thinking compensation mechanisms for authors of copyrighted works that are ingested as part of model training.

Additionally, it is important to note that technological innovations toward mitigating downstream copyright infringements–and, perhaps, technological harms in general–do not come only from strict compliance with legislation. They are heavily influenced by organizational efforts and informal standards setting [Menell 2012]. There are multiple policy avenues–standards setting, private ordering–through which technological innovation in preventing infringement can be moved. To solidify and develop these duties and standards, partnerships with the open-source ML community, which increasingly shapes the basis of ML development, may play a central role [Langenkamp and Yue 2022]. Standards-setting through governmental bodies, such as the National Institute of Standards and Technology and the Federal Trade Commission, is another way that regulators can shape the incentives that determine AI development.

Finally, copyright law is by no means a panacea for regulating AI development and deployment or for protecting artists in that process. The burden of proof currently falls on artists to discover works that might have infringed on theirs and to bring an action. The resource-intensive process could dissuade many lawsuits from being brought in the first place [Choi 2023]. In light of this, it may be important to assess whether copyright fully captures the intellectual property rights that are needed to defend incentives for creative work. The seminal copyright law was developed in response to a new technology—the printing press—that could produce cheap copies of an author's work [8 Anne, c. 19 (1710); Mills 1989]. Although the reproduction of copyrighted works through ML systems is important, many current concerns relate to the ability of model developers to use copyrighted works as a means to compete, directly or indirectly, with the people who produced that data. A novel feature of generative models, which distinguishes them from technologies such as the printing press, which produce exact copies, is that they do not need to reproduce or plagiarise an artist's work to interfere with the market for the artist's *skills*. In this sense, it may make sense to think of these models as interfering with the market for the *creator* in addition to the market for the *work*. This raises questions about what rights are needed to preserve incentives for people to contribute to the "Progress of Science and the useful Arts" (U.S. Const. art. I, §8, cl. 8) in response to the development of machine learning and AI systems.

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